# TerraMind-Tiny

## Satellite SAR Ship Detection for Maritime Law Enforcement

418 Hackathon by Enigma

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# The Market: Ghost Ships & Real Money

## The Problem

Ships disable AIS to evade tracking—enabling illegal fishing, smuggling, and sanctions evasion in international waters.

## Who Wants This Data?

### 🏛 Government

Maritime law enforcement, border security, sanctions compliance

### 💰 Traders & Hedge Funds

Real-time commodity intel. See oil barrels moving BEFORE market knows. 10-15min edge = millions in profit. Higher margins than government can provide from the data.

# Our Solution: Lean & Efficient

## TerraMind-Tiny Architecture

✓ LoRA fine-tuning (r=1)

✓ VH/VV SAR modalities

✓ Custom detection head

### Why No Thinking in Modalities?

TiM generates modality context—costs extra RAM/VRAM storage. For orbital compute with strict memory limits, we analyzed the tradeoff: minimal gain vs significant overhead.

**Result: Direct VH/VV fusion. Simpler, faster, edge-compatible.**

Model Size

**21.09 MB**

*3x smaller*

Throughput

**2,151 samples/sec**

*Real-time*

Parameters

**5.4M**

*Efficient*

Target

**Orbital Hardware**

*Edge-ready*

# Performance: TerraMind vs Baseline

| Metric | TerraMind (Ours) | Baseline (YOLOv8m) | Impact |
|---|---|---|---|
| Accuracy | **73.61%** | 63% | +10.6% |
| Precision | **81.93%** | — | High confidence |
| F1 Score | **80.41%** | — | Balanced |
| Model Size | **21 MB** | 30 MB | 3x smaller |

**Test Set: 720 samples | True Positives: 390 | True Negatives: 140**

# What Works. What Doesn't. What's Next.

## Current Limitations

⚠ AIS data currently historical/purchased (not real-time public)

⚠ High sea-state conditions reduce accuracy (wave clutter)

⚠ Generalization limited to Sentinel-1 sensor

## Next Priorities

✓ Integrate live public AIS data streams

✓ Expand to multi-sensor SAR imagery

✓ Robust performance in extreme weather

✓ Deploy to orbital edge hardware